Furthermore, the wireless world is seeing the regular birth of new technologies, with more on the horizon. Some of these technologies will expand the functionality of the device while others will dramatically change their connectivity with other devices (e.g., Bluetooth technology).

No users have lost data as a result of Palm Phage and the EPOC joke programs. But this malicious code ups the ante for such code in the wireless arena—demonstrating that self-replicating viruses are not only possible to develop, but easy to develop. And with the expanded functionality of these devices in the coming months and years, so will expand the potential for new threats from malicious code.

Content-based Threats

In content-based threats, the content (e.g., derogatory messages) is the threat, or malicious use of the content is the threat (e.g., spamming of e-mail). While e-mail has become the 'killer app' of the wireless world, it is also one of the most vulnerable to attack. Hence, the most common content-based threats to the wireless infrastructure occur through infected e-mail or spam mail.

The first content-based Trojan to attack wireless devices occurred in June 2000 with the appearance, in the wild, of the Visual Basic Script (VBS) Timofonica on the wireless network of Madrid, Spain-based Telefonica SA. Timofonica spread by sending infected e-mail messages from affected computers. When an infected e-mail reached a PC, it used Microsoft Outlook 98 or 2000 to send a copy of itself via infected e-mails to all addresses in the MS Outlook Address Book. This enabled the Trojan to spread quite rapidly. In the wired world, this behaviour is similar to that of the "ILoveYou" e-mail virus that caused worldwide damage estimated as high as $700 million in May 2000.

But Timofonica was more than an e-mail virus. For each e-mail it sent, the Trojan also dispatched an SMS message to a randomly generated address at the "correo.movistar.net" Internet host (see Figure 4). Since this host sends SMS messages to mobile